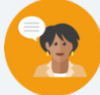


12.6 Estimating for a Population

Lesson Introduction

 Benchmark	MA.912.DP.1.4 Estimate a population total, mean, or percentage using data from a sample survey; develop a margin of error through the use of simulation.		 Learning Target	I can use margin of error to estimate a population total, mean, or percentage using data.
 Benchmark Coherence	Building On		Working Toward	
	MA.7.DP.1.3 Given categorical data from a random sample, use proportional relationships to make predictions about a population.		MA.912.DP.1.5 Interpret the margin of error or percentage from a data set. Interpret the confidence level corresponding to the margin of error.	
 Essential Question	How can I use margin of error to estimate a population total, mean, or percentage using data?			
 Lesson Narrative	In this lesson, students are introduced to the concept of margin of error and learn how to estimate for a population. Students will receive a formal introduction to margin of error, and use margin of error to make estimates for a population.			
 Component Summary	12.6.1 Warm-Up (~5 min.)	News source Notice and Wonder (<i>SE p. 225</i>)	12.6.3 Guided Practice (10-15 min.)	Practice estimating for a population (<i>SE p. 227</i>)
	12.6.2 Guided Instruction (10-15 min.)	Introduction to margin of error (<i>SE p. 225</i>)	12.6.4 Your Turn! (10-15 min.)	Independent practice with margin of error and estimating for a population (<i>SE p. 228</i>)
	12.6.5 Wrap-Up (~5 min.)	Unit concepts reflection (<i>SE p. 229</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> Margin of error <u>Additional Terms:</u> N/A		N/A	Work through this lesson prior to facilitating with students. It builds a conceptual understanding of margin of error related to the population.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not recognize the role of the margin of error without prompting from the question stem. - Students may calculate the values incorrectly. - Students may draw incorrect conclusions about the population.	The most important part of this lesson is for students to conceptually understand a margin of error and its role in estimating information about a population. Continue to focus in on that conceptual piece through each part of the lesson.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 12.6.1 Warm-Up provides students with an opportunity to review a graphic that summarizes data about website and news sources through a Notice and Wonder instructional routine. The use of this routine makes the task accessible for all students.	MA.K12.MTR.7.1: Real-World Contexts Throughout this lesson, students will encounter problems that represent real-world contexts. Students will be able to connect these mathematical concepts to everyday experiences.
 Differentiate Instruction	Consider creating a visual reference for the concept of margin of error. This should include not simply a definition, but 2-3 examples of it being used to provide information about the overall population.	
Lesson Notes:		

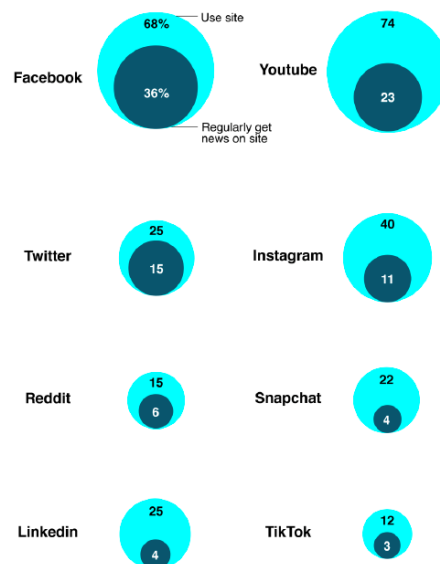
12.6.1 Warm-Up: News Source

Component Overview: Students will review a graphic about news sources to articulate what they notice and what they wonder about the data.



12.6.1 Warm-Up: News Source

- The graphic below summarizes data from a 2020 Pew Research Study. It shows the percent of U.S. adults who use each site and regularly get news on each site.



What do you notice? What do you wonder?

Answers will vary. With the exception of Facebook and Twitter, the percentage of adults who regularly get news from each site is less than half of the percentage of adults who use the site. However, those two platforms have high usage, so it is still a large portion overall receiving news from those sites.



12.6.1 Warm-Up provides students with an opportunity to review a graphic that summarizes data about website and news sources through a Notice and Wonder instructional routine. The use of this routine makes the task accessible for all students.

12.6.2 Guided Instruction: Margin of Error

Component Overview: Students will be introduced to the concept of margin of error and what it means to the population for the given context.



12.6.2 Guided Instruction: Margin of Error

In 2020, 9,220 U.S. adults were surveyed about their news use across social media platforms.

53% of those surveyed reported that they get their news from social media “often” or “sometimes.”

1. If you were able to conduct a similar survey of U.S. adults, do you think exactly 53% would report that they get their news from social media “often” or “sometimes?” Explain your reasoning.

Answers will vary. No, it is unlikely that a similar survey would produce the exact same results as the survey in this scenario. Different people could be surveyed, the survey would be happening at a different time, the survey could happen in a different region of the U.S. These and other factors could contribute to different results.

2. Can it be concluded that exactly 53% of all U.S. adults get their news from social media “often” or “sometimes?”

Answers will vary. No. This statistic is true of the specific people surveyed from the specific time they were questioned. If the sample of people surveyed was selected in a way to prevent bias, then the results should be roughly accurate and true for other groups of U.S. adults, but not exact.



It is important that students conceptually understand a margin of error and its role in estimating information about a population. Continue to focus in on that conceptual piece throughout, using consistent and appropriate vocabulary so students can connect it to the context of the problems.

12.6.2 Guided Instruction: Margin of Error (continued)

3. Do you think that the majority of U.S. adults get their news from social media sometimes or often?

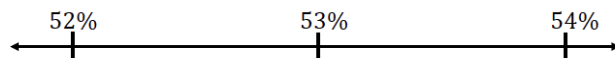
Answers will vary. It is possible that a majority of U.S. adults get their news from social media sometimes or often, but when the results of both studies are combined, this seems less likely. Since 53% of U.S. adults surveyed got their news from social media sometimes or often, which is just slightly over half, and a vast majority of the data included in the Pew Research Study show less than half of those surveyed get their news from social media sites they use often, it is unlikely that most, or over half, of adults in the U.S. get their news from social media sometimes or often.

4. Estimate the total number of U.S. adults who got their news from social media often or sometimes in 2020 if there were 256,662,010 U.S. adults at the time of the survey.

$0.53 \cdot 256,662,010 = 136,030,865.3$ *This means that an estimate of the total number of U.S. adults who got their news from social media often or sometimes in 2020 is 136,030,866.*

When a population percentage or mean is estimated from sample data, it is often helpful to give a range for the estimate to consider variability in the sample.

Based on the same sample data, it can be concluded that the population percentage is between 52% and 54%.



This interval of possible values for the estimate of the population percentage was created using a **margin of error**.

The **margin of error** quantifies the amount of random sampling error in the results of a survey.



Why do you think it can be concluded that the population percentage is between 52% and 54%? Let students observe that the sample percentage, 53% is in the middle of the range given.

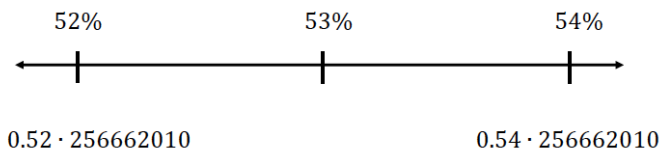


Consider creating a visual reference for the concept of margin of error. This should include not simply a definition, but 2-3 examples of it being used to provide information about the overall population.

12.6.2 Guided Instruction: Margin of Error (continued)

In the sample data of 9,220 U.S. adults, the percentage who get their news from social media "often" or "sometimes" is 53% with a margin of error of $\pm 1\%$.

5. Estimate the range for the total number of U.S. adults who got their news from social media often or sometimes in 2020 if there were 256,662,010 U.S. adults at the time of the survey using the estimate of 53% with a margin of error of $\pm 1\%$.



The total number of U.S. adults who get their news from social media sometimes or often is estimated to be between 133,464,245 and 138,597,486.



Consider providing students with an opportunity to discuss why this might be problematic, or what implications or risks come along with this data. Guide students to not just finding the results, but also thinking through the implications of their findings.

12.6.3 Guided Practice: Estimating for a Population

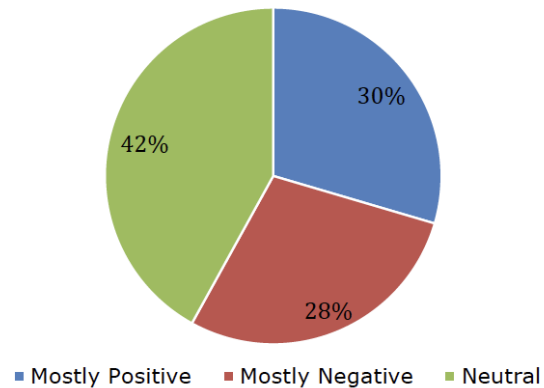
Component Overview: Students will practice, using three real-world questions, estimating for a population.



12.6.3 Guided Practice: Estimating for a Population

1. The results of 741 teens surveyed in a 2018 Pew Research Study are shown in the circle graph.

Teen Views on the Effect of Social Media on People Their Age



- a. If there were 24.97 million teens in the U.S. population at the time of the survey, what is the estimated number who believed the effect of social media on people their age had been mostly positive?

$$24.97 \text{ million} \cdot 0.30 = 7.49 \text{ million teens}$$



Consider allowing students to work in partner pairs to try each question, as some have discussion components that can provide for a rich, mathematical discussion among peers.

12.6.3 Guided Practice: Estimating for a Population (continued)

- b. The proportion of all U.S. teens who believe the effect of social media has been neutral on people their age is 0.42 with a margin of error of 0.036. Estimate the range of all U.S. teens who believe the effect of social media has been neutral on people their age.

First find the lower possible proportion using the margin of error: $0.42 - 0.036 = 0.384$.

Next use this to calculate the low end of the estimate: $0.384 \cdot 24.97 = 9.59$.

Now find the higher possible proportion using the margin of error: $0.42 + 0.036 = 0.456$.

Then use that to find the high end of the estimate: $0.456 \cdot 24.97 = 11.39$.

This means an estimate of the number of all U.S. teens who believe the effect of social media has been neutral on people their age is between 9.59 million and 11.39 million.

2. In a survey of 175 students at Satellite High School (SHS), 82 reported that they use Facebook.
- a. The estimated proportion of all Satellite High School students who use Facebook is 0.469 with a margin of error of 0.074.
- b. The estimated proportion of all Satellite High School students who use Facebook is between 0.395 and 0.543.
- c. Do you think it's reasonable to conclude that the majority of SHS students use Facebook? Explain your reasoning.

Answers will vary. No, I do not think it's reasonable to conclude that the majority of SHS students use Facebook because the range of estimates for the proportion of students who use Facebook is 0.395 to 0.543. This range includes values below 0.50, so it is not reasonable to say that the majority of SHS students use Facebook.



How did you determine the margin of error? What percentage constitutes a majority?

12.6.3 Guided Practice: Estimating for a Population (continued)

- d. Suppose there are 1,385 students enrolled at Satellite High School. The estimated number of all Satellite High School students who use Facebook is between 547 and 753.

3. A large company tracked the amount spent on each of their Amazon purchases over the past four years. For a sample of 135 purchases, the average amount spent was \$35.23 with a margin of error of $\pm \$2.16$.

If the company predicts they will make 50 purchases on Amazon in the next month, what is the range of the predicted total amount they will spend?

First calculate the range of the average amount spent per purchase:

$$\$35.23 - \$2.16 = \$33.07 \text{ and } \$35.23 + \$2.16 = \$37.39.$$

Next multiply both by 50 to find the range of the predicted total amount they will spend:

$$\$33.07 \cdot 50 = \$1,653.50 \text{ and } \$37.39 \cdot 50 = \$1,869.50.$$

The range of the predicted total amount they will spend on Amazon in the next month is \$1,653.50 to \$1,869.50.



How can knowing this information be beneficial for a company?

12.6.4 Your Turn!

Component Overview: Students will practice three real-world questions independently to estimate for a population.



12.6.4 Your Turn!

1. A new drug was given to a random sample of 400 people with acne. The results showed that 210 of the people's acne improved and the results had a margin of error of ± 0.049 .

- a. Based on these results, can it be said that more than half the people with acne will improve if they are given the drug?

Answers will vary. No. While it's true that over half of the random sample, 52.5%, saw their acne improve during the time they took the drug, the margin of error is 4.9%. When the margin of error is applied the percentage dips to 47.6%. Since this is below half, it is not a good idea to claim that more than half of the people with acne that take the drug will see an improvement in their acne. A better way to summarize the results of the study would be to say that just over half of the people in the study saw an improvement in their acne while taking the drug; however, there was a margin of error of almost 5%.

- b. If the new medication will be administered to 5,552 people, how many are expected to see improvement?

Between 2,642 and 3,187 are predicted to see an improvement in their acne.



Consider using this time to work with students who may need additional support on this topic based on their performance during the prior components.

12.6.4 Your Turn! (continued)

2. During harvesting season at Florida Blue Farms, hand pickers collected 200 pounds of blueberries and found a 95% pack-out rate. The margin of error for the sample was $\pm 3\%$. The pack-out rate is the percentage of collected blueberries that can be packaged to be sold, based on Florida Blue Farms' quality standards.
 - a. The estimated percentage of all blueberries that are hand-picked and can be packaged and sold is between $95\% - 3\% = 92\%$ and $95\% + 3\% = 98\%$.
 - b. If 2,800 blueberries are hand-picked, what is the most that are expected to be packaged and sold?
 $2,800 \cdot 0.98 = 2,744$
If 2,800 blueberries are hand-picked, the most blueberries that are expected to be sold is 2,744.
3. Publix Supermarkets Inc., commonly known as Publix, has 831 locations in Florida. Publix offers free cookies to children to eat while they're in the store. A survey of 100 store managers found that the average number of cookies given out on a weekday was 15.5, with a margin of error of 2.1.

Complete the statement.

The estimated total number of free cookies given out each weekday for all Florida locations is between

$831(15.5 - 2.1) = 11,135$ and $831(15.5 + 2.1) = 14,626$.



What is the least number of blueberries that could be sold?



If each store gave out the same amount, what would be the total number of cookies given out at a single store each weekday?

12.6.5 Wrap-Up: Reflection

Component Overview: Students complete a self-reflection on their level of understanding of the concepts from this unit.



12.6.5 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each.

Answers will vary.



I need help.



I got it.



I could help someone.

Classifying data as categorical or quantitative, and as univariate or bivariate	  
Creating and interpreting graphical displays for categorical data	  
Creating and interpreting graphical displays for numerical data	  
Explaining the difference between correlation and causation	  
Using the margin of error to estimate population totals, means, and percentages	  



Consider using data from this formative assessment to determine which students need a review of which learning targets prior to the end of unit assessment.